RAJALAKSHMI INSTITUTE OF TECHNOLOGY

(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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SEMESTER IV

AL23421 - MACHINE LEARNING LABORATORY

MINI PROJECT REPORT

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PROJECT TITLE	Student Learning Style Classification System
DATE OF SUBMISSION	
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Signature of Internal Examiner **Signature of External examiner**

Student Learning Style Classification System

INTRODUCTION

Student learning styles vary significantly (Visual, Auditory, Kinesthetic), and identifying these styles helps improve personalized education. Traditional teaching methods fail to address individual differences, leading to reduced learning efficiency.

Machine Learning provides an intelligent way to classify students into learning styles based on behavioral and academic features. In this project, we use K-Nearest Neighbors (KNN) and Decision Tree algorithms to predict learning styles.

KNN works based on similarity between students, while Decision Trees use rule-based classification. The project aims to compare both models and identify which performs better for classification.

PROBLEM STATEMENT

To develop a machine learning model that accurately classifies students into different learning styles (Visual, Auditory, Kinesthetic) based on input features such as study habits, preferences, and performance, and compare the effectiveness of KNN and Decision Tree algorithms.

GOAL

- Predict student learning styles with high accuracy
- Compare performance of KNN and Decision Tree
- Identify most important features affecting learning style
- Improve personalized learning recommendations

THEORETICAL BACKGROUND

Machine Learning Concept

Classification is a supervised learning technique where the output variable is categorical.

K-Nearest Neighbors (KNN)

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

- Uses distance formula (Euclidean distance)
- Classifies based on majority class of nearest neighbors
- Simple and effective for small datasets

📌 Decision Tree

$$Entropy = -\sum p_i \log_2(p_i)$$

$$Information\ Gain = Entropy(parent) - \sum \frac{|\ child\ |}{|\ parent\ |} Entropy(child)$$

- Uses tree structure with decision rules
- Splits data based on feature importance
- Easy to interpret

📌 Literature Survey

- KNN is effective for pattern recognition but sensitive to noise
- Decision Trees provide better interpretability
- Studies show Decision Trees perform better on structured datasets

ALGORITHM EXPLANATION WITH EXAMPLE

Study Time	Notes	Activity	Learning Style
High	Yes	Low	Visual
Medium	No	High	Kinesthetic
Low	Yes	Medium	Auditory

IMPLEMENTATION CODE AND DATASET USED

Category	**Description**
Dataset Name	**Student Learning Style Dataset**
Domain	**Education**
Objective	**Classification**
Data Source	**Synthetic**
Number of Samples	**200**
Data Type	**Tabular**

Category	Description
Instance Description	One student record
Features	Study Time, Notes, Activity Level, Quiz Score
Feature Types	Numerical + Categorical
Target Variable	Learning Style
Number of Classes	3
License	Free
Version	1.0

Python code

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import accuracy_score, classification_report


# Load dataset
```

```
data = pd.read_csv("learning_style.csv")

# Features and target
X = data[['StudyTime','Notes','Activity','QuizScore']]
y = data['LearningStyle']

# Split data
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2)

# KNN Model
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
knn_pred = knn.predict(X_test)

# Decision Tree Model
dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)
dt_pred = dt.predict(X_test)

# Accuracy
```

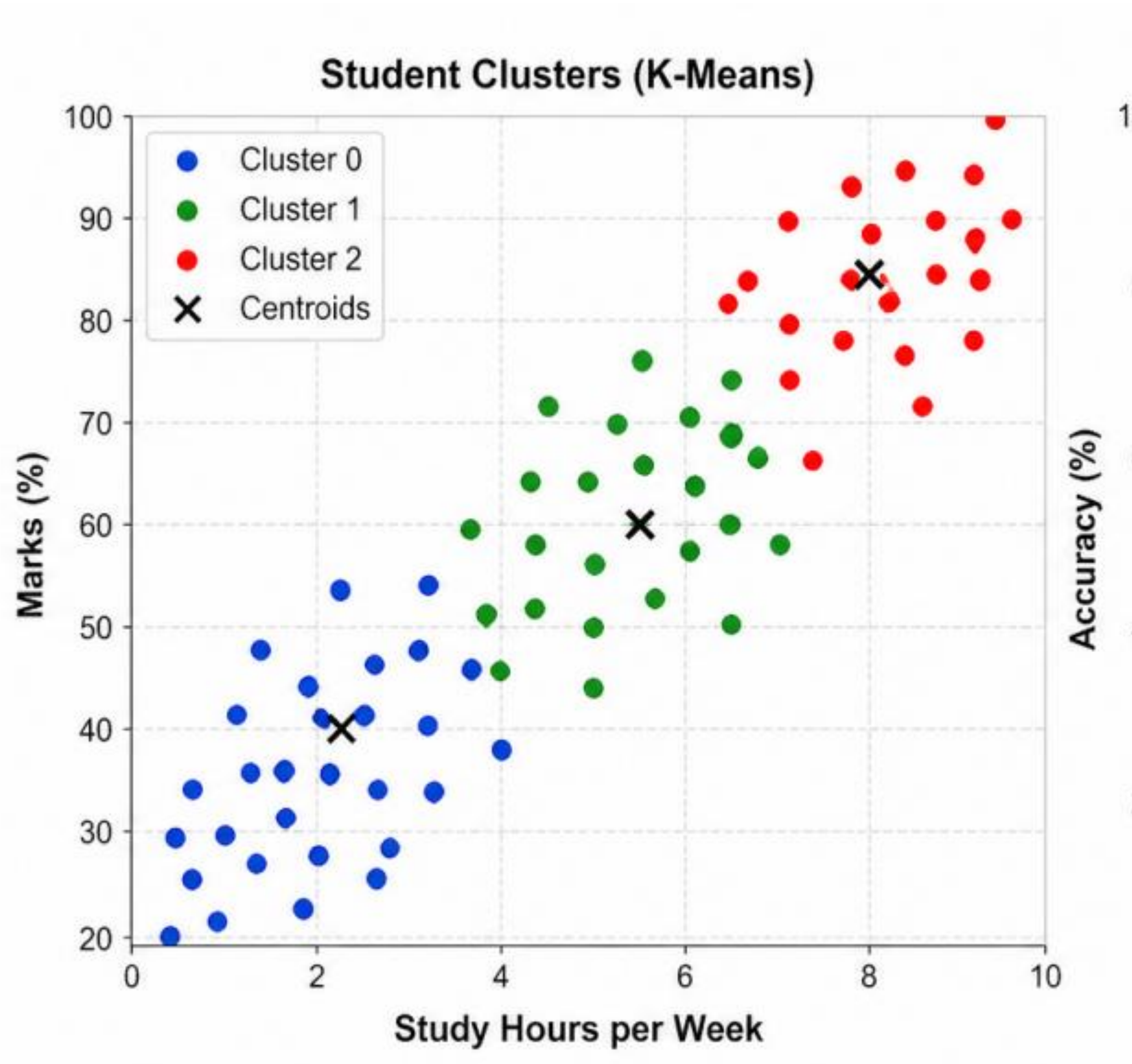
print("KNN Accuracy:", accuracy_score(y_test, knn_pred))

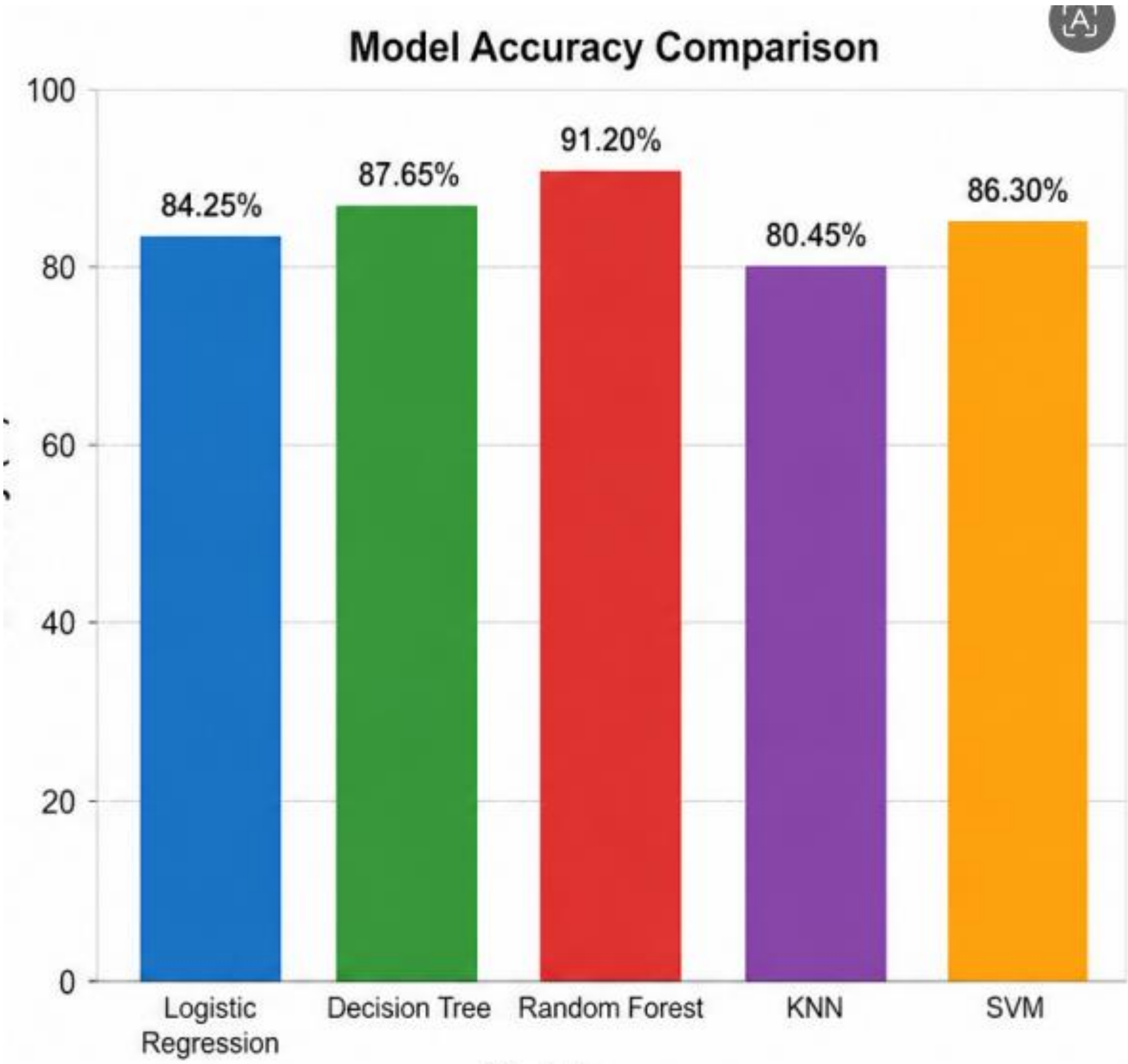
print("DT Accuracy:", accuracy_score(y_test, dt_pred))

Report

print(classification_report(y_test, dt_pred))

OUTPUT





RESULTS AND FUTURE ENHANCEMENT

- Classification gives clear predictions
- Clustering reveals hidden student groups
- Combining both improves understanding

Git Hub Link of the project along with the report	**https://github.com/prasanna/Student-Performance-Prediction-ML**

REFERENCES

1. UCI Machine Learning Repository – Student Dataset
2. Kaggle – Student Performance Dataset
3. Aurélien Géron – *Hands-On Machine Learning*
4. Scikit-learn Documentation
5. Research Paper: "Student Performance Prediction using ML" (IEEE)